OUT “5000” & Amount to be returned DIV 5000

Amount to be returned = Amount to be returned MOD 5000

Programming Fundamentals Lab

NAME: MUHAMMAD HASSAN KHALID

ROLL NO: 21L-5692

FIRST ACTIVITY

ALGORITHM NUMBER 1

10 We have to create two variables Radius and Area

20 Get a value of radius by the user

30 Calculate area by using the formula given below

Area = 3.1415 * Radius * Radius

40 Print the value of area on the screen

ALGORITHM NUMBER 2

10 We have to create three variables width, breadth and area

20 Get values of width and breadth by the user

30 Calculate area by using the formula given below

Area = Width * Breadth

40 Print the value of area on the screen

ALGORITHM NUMBER 3

10 We have two create two variables Number1 and Number2

20 Get values of Number1 and Number2 by the user

30 Compare the two numbers and see which is larger and print in ascending order

If Number1 > Number2 then

Output Number1

Output Number2

Else

Output Number2

Output Number1

EndIF

ALGORITHM NUMBER 4

10 We have to create four variables Count5000, Count1000, Count500 and Total

20 We have to count the number of note available in the machine

Repeat

If Note=5000 then Count5000=Count5000 + 1

If Note=1000 then Count1000=Count1000 + 1

If Note=500 then Count500=Count500 + 1

Until Note<>0

30 Calculate Total

Total=Count5000 + Count1000 + Count500

40 Output Total on ATM Machine

ALGORITHM NUMBER 5

10 We have to create four variables Final Velocity, Initial Velocity, Acceleration and Time

20 Get the value of time acceleration and initial velocity by the user

30 Calculate final velocity using the formula given below

Final Velocity = Initial Velocity + Acceleration*Time

40 Output Final Velocity on the screen

ALGORITHM NUMBER 6

10 We have to create two variables Radius and Circumference

20 Get a value of radius by the user

30 Calculate circumference by using the formula given below

Circumference = 2 * 3.1415 * Radius

40 Print the value of circumference on the screen

ALGORITHM NUMBER 7

10 We have two create two variables Number1 and Number2

20 Get values of Number1 and Number2 by the user

30 Compare the two numbers and see which is smaller and print in descending order

If Number1 < Number2 then

Output Number1

Output Number2

Else

Output Number2

Output Number1

EndIF

ALGORITHM NUMBER 8

10 Take two variables length and area

20 Get value of length by the user

30 Calculate area

Area = 5 * Length * Length

40 Output Area on the screen

Algorithm Number 9

10 Take two variables Celsius and Fahrenheit

20 Calculate Celsius temperature

Celsius = (9/5 * Fahrenheit) + 32

30 Output Celsius

Algorithm Number 10

10 Take two variables Celsius and Fahrenheit

20 Calculate Fahrenheit temperature

Fahrenheit = (Celsius – 32) * 5/9

30 Output Fahrenheit

Second Activity

Part A

Algorithm Number 3

10 CR W,B,A

20 IN W,B

30 A =W*B

40 OUT A

Algorithm Number 5

10 CR IV,FV,A,T

20 IN IV,A,T

30 FV = IV + A*T

40 OUT FV

Algorithm Number 6

10 CR C,R

20 IN R

30 C = 2*3.1415*R

40 OUT C

PART B

PROBLEM NO 1

CR N, B, Number Boys, Total Children, Number Girls, Percentage B, Number Girls B, Number Boys B

IN N, B

Number Boys = (B * Total Children) / 100

Number Girls = Total Children - Percentage Boys

Percentage B = (Total Children * 30) / 100

Number Boys B = (Percentage B * 40) / 100

Number Girls B = Percentage B - Number Boys B

OUT Number Boys, Total Children, Number Girls, Percentage B, Number Girls B, Number Boys B

PROBLEM NO 2

IN Total Bill, Amount Paid, Amount to be returned

IF Amount Paid < Total Bill Then

OUT “Less Amount Paid”

Else

Amount to be returned = Amount Paid – Total Bill

OUT “Amount” & Amount to be returned

OUT “5000” & Amount to be returned DIV 5000

Amount to be returned = Amount to be returned MOD 5000

OUT “1000” & Amount to be returned DIV 1000

Amount to be returned = Amount to be returned MOD 1000

OUT “500” & Amount to be returned DIV 500

Amount to be returned = Amount to be returned MOD 500

OUT “100” & Amount to be returned DIV 100

Amount to be returned = Amount to be returned MOD 100

OUT “50” & Amount to be returned DIV 50

Amount to be returned = Amount to be returned MOD 50

OUT “20” & Amount to be returned DIV 20

Amount to be returned = Amount to be returned MOD 20

OUT “10” & Amount to be returned DIV 10

Amount to be returned = Amount to be returned MOD 10

OUT “5” & Amount to be returned DIV 5

Amount to be returned = Amount to be returned MOD 5

OUT “2” & Amount to be returned DIV 2

Amount to be returned = Amount to be returned MOD 2

OUT “1” & Amount to be returned DIV 1